

USDA Interviews

Micaela Ciambrone, Layilah Campbell, Seorin Jang, Dongsub Kim, Shenghao Mei

Harvesting Insights: Usability Study of the ACIR Database

SI 622

THE TEAM



Micaela Ciambrone



Seorin Jang



Layilah Campbell



Sam Mei



Aiden Kim

Introduction:

Purpose:

Evaluate how effectively users navigate, interpret, and apply regulatory information in the Agricultural Commodity Import Requirements (ACIR) portal.

Project Focus:

- Assess navigation, search, and filter usability
- Evaluate clarity and organization of help resources
- Examine mobile experience and small-form-factor use cases

Key Research Areas:

- Efficiency of upfront vs. post-search filtering
- User ability to locate commodity-specific requirements
- Effectiveness of current help pathways (labels, videos, guides)
- Opportunities for a centralized “single help hub”



ACIR Search Tools - Locator Table and Frequently Searched

Are you looking for a way to browse topics in ACIR? Would you like a prebuilt search list of Commodity Import Requirements (CIRs) on a specific topic?

There are two Search Tools available in ACIR:

- **Locator Table** - Found on the [ACIR Help Page](#)
 - Allows users to browse various commodities and locate specialty items in ACIR.
 - Topics are listed alphabetically and include Bags/Bagging, Bee Products, Corn, Fresh Fruits and Vegetables, Herbarium Specimens, Lichen, Live Plants, Plant Oils, Seed Kits, Wooden Handicrafts and much more.
- **Frequently Searched** - Found on all search tile pages, above the Search button
 - Allows users to choose prebuilt searches for quick and easy results based on the tile content.
 - [Plants and Plant Products Not for Propagation](#) has the most topics available (e.g., Seeds, Nuts, and Grains - Fresh Fruit and Vegetables from a certain country - Handicrafts - Mushrooms - Tea Products, etc.)
 - Click on the Select button to the right of the topic to view the search results with commodity import requirements related to the topic.

Methods

01

Recruitment

Participants were recruited directly by USDA, though specific selection criteria were not disclosed.

02

Participants

Seven USDA employees from multiple departments and states participated, providing cross-functional and geographic perspectives on ACIR usage.

03

Data Collection

We conducted 45–60 minute semi-structured interviews exploring how professionals locate, interpret, and apply agricultural import regulations, with recordings transcribed and anonymized.

04

Analysis

Transcripts were independently coded by the interviewer and notetaker to identify recurring themes, contradictions, and cross-departmental patterns that informed our findings.

Key Findings/Recommendations

- **Search & Navigation Barriers:** Users frequently overlook the "Control-F" capability within PDF viewers and have permanently abandoned search filters due to early technical frustrations.
- **Data Misclassification:** Genus-level tagging causes "herbaceous" seeds to be mislabeled as "woody," leading to incorrect regulatory treatment by CBP.

- **Tile Misinterpretation:** Commercial importers frequently misunderstand the "Pest Organism" tile, which is intended solely for specialized research and medical communities.
- **Mobile & Field Utility:** Field Ops personnel lack a optimized mobile experience, struggling with sizing issues and features that are difficult to use outside of a dedicated app.
- **Communication Gaps:** There is significant confusion regarding "theoretically possible" but rarely approved permits, such as those for African Snails

Recommendations

- **Formal Retraining:** Prioritize user education on PDF searching and filter functionality over further system changes.
- **Enhanced Signaling:** Use high-priority visual cues to clarify the niche research intent of the "Pest Organism" tile.
- **Manual Notification:** Implement a case-by-case manual process to resolve woody vs. herbaceous classification errors.
- **Field Optimization:** Refine the responsive web design to better support handheld device use during port inspections.
- **Expectation Management:** Improve instructional text to clarify the low likelihood of approval for high-risk permits.

Meet Marcus Bennett:

Marcus Bennett



ROLE

Operations Manager,
Agricultural Import
Company

“ The database gives me the requirement, but the manual gave me the context.

Primary Goal

Determine quickly whether a product can be imported and what documentation is required.

Background

Marcus manages logistics and supplier coordination for a mid-sized seed and plant distributor. He's business-savvy but not trained in regulatory interpretation. He previously struggled with dense PDF manuals and avoids complex filtering systems because they feel overly technical.

How She Uses ACIR

- Searches specific commodities
- Looks for simple admissibility confirmation
- Skips advanced filters

Frustrations

- Regulatory language is dense
- Filtering system feels confusing
- Too many clicks to reach an answer

Core Needs

- Plain-language summaries
- Guided decision pathway
- Clear checklist of required steps

Meet Angela Rivera:

Angela Rivera



ROLE

Trade Director,
International
Agricultural Policy

“ The database gives me the requirement, but the manual gave me the context.

Primary Goal

Ensure agricultural trade between countries complies with U.S. regulations while maintaining clarity and consistency across stakeholders.

Background

Angela has spent over a decade working in international agricultural trade. She frequently communicates with foreign regulatory agencies and exporters who rely on ACIR to understand U.S. import requirements. She understands regulatory language deeply and often compares current ACIR entries with legacy manuals.

How She Uses ACIR

- Searches by country and commodity
- Cross-checks unclear entries against older manuals
- Clarifies ambiguous requirements for trade partners

Frustrations

- ACIR feels too siloed by commodity and country
- Cannot search by general regulatory concept (e.g., definitions across commodities)
- Feels some nuance was lost when manuals were consolidated

Core Needs

- Concept-based search
- Linked regulatory definitions
- Better traceability from legacy manual language

Meet Claire Thompson:

Claire Thompson



ROLE PPQ Officer, Field Operations (Plant Inspection Station)

“I just need it to clearly say yes or no, and why.

Primary Goal

Quickly determine admissibility requirements and provide accurate guidance to brokers and importers.

Background

Claire works on the front lines reviewing imports daily. She makes regulatory decisions that directly impact shipments at ports of entry. She is highly familiar with ACIR and uses it regularly, often under time pressure.

How She Uses ACIR

- Searches commodity and country combinations
- Cross-checks taxonomy and admissibility tiles
- Interprets seed limits and treatment requirements

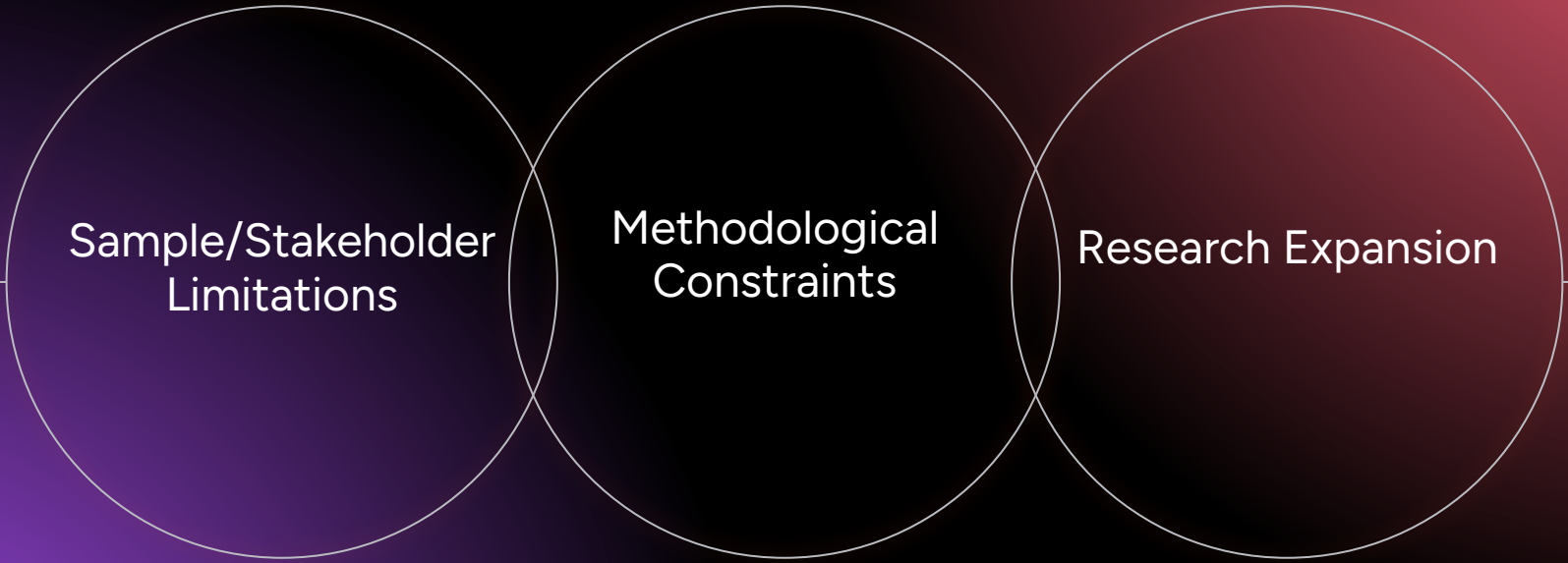
Frustrations

- Inconsistencies between tiles (e.g., taxonomy vs. admissibility)
- Ambiguity in wording (e.g., “small lot of seeds” limits)
- Must explain nuanced regulatory language to non-experts

Core Needs

- Conflict alerts between tiles
- Embedded clarifications and definitions
- Real-world examples within entries

Discussions/Future of Work



Sample/Stakeholder
Limitations

Methodological
Constraints

Research Expansion

THANK YOU

Any questions?